Microeconomics

The consumer problem

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From preferences to an actual choice

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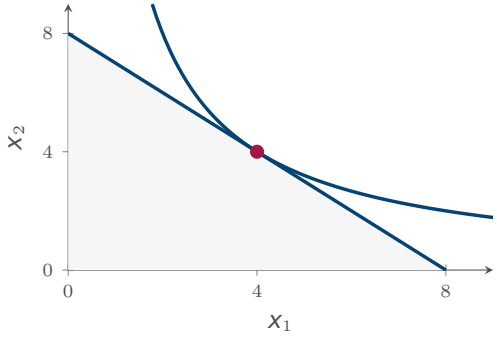
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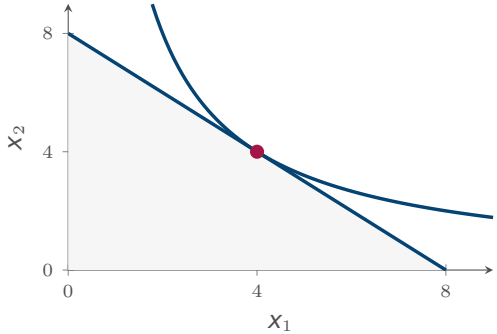
A separate reserve section introduces expenditure minimisation and Hicksian demand for the next course block.

The picture we already know



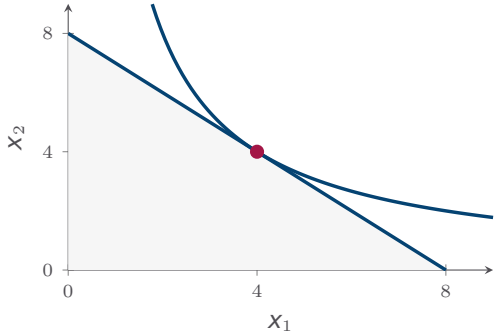
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$u(x) = \sqrt{x_1 x_2}$, $p = (1, 1)$, $m = 8$. Which affordable bundle is best? $(4, 4)$. Today we want a systematic method, not just a convincing picture.

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Preferences enter through u; prices and wealth enter through the feasible set.

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What changes for a subscription, a quantity discount or an indivisible purchase?

Two useful facts about the budget

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Nonemptiness: the zero bundle is affordable.
Compactness: the budget is closed, and $0 \leq x_i \leq m/p_i$ for every i.
Strictly positive prices give a finite upper bound on every quantity.

Does an optimum exist?

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Existence does not yet imply uniqueness, an interior solution or differentiability of demand.

Which assumption is doing the work?

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Can you give a case with multiple maximisers even though the budget is compact?

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NES

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It is the earlier choice rule applied specifically to budget menus:
$D(p, m) = C(B(p, m))$.

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This is a change of currency units, not an increase in purchasing power.

Budget exhaustion: Walras' law

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Therefore an existing optimum spends the budget. → Proof

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Convex preferences make z_t at least as good as every such bundle too.
Thus $D(p, m)$ is a convex set. This still allows an entire line segment of solutions.

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→ Proof Does strict convexity by itself prove existence?

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Demand tells us **which bundle**; indirect utility tells us **the attained value**. Even if $D(p, m)$ contains several bundles, every one gives the same value $v(p, m)$. It is a function of the environment, not of a chosen bundle.

How the optimal value changes

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Wealth: under LNS, raising m strictly increases v.

Continuity: continuous u gives continuous v on the maintained positive-price, positive-wealth domain.

A less obvious property: quasiconvexity

NES

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If a bundle is unaffordable at both endpoints, it is unaffordable at their mixture too.

So the mixed budget cannot deliver utility above both endpoint values. → Proof

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Smooth interior problem: marginal conditions can identify a candidate.

Always check feasibility, boundary cases and a reason why the candidate is globally optimal.

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If both goods are consumed in a smooth optimum, these marginal returns must agree.

The Lagrangian: one multiplier per inequality

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For the smooth argument, assume u is differentiable on an open neighbourhood of the candidate.

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Stationarity is only one group of conditions. We must also check signs, feasibility and slackness.

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Stationarity: $u_i(x^*) - \lambda^* p_i + \mu_i^* = 0.$
Primal feasibility: $p \cdot x^* \leq m$, $x^* \geq 0.$
Dual feasibility: $\lambda^* \geq 0$, $\mu^* \geq 0.$
Complementary slackness:

$$\lambda^*(m - p \cdot x^*) = 0, \qquad \mu_i^* x_i^* = 0.$$

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If a multiplier is strictly positive, its constraint must bind.
Does a binding constraint always have a strictly positive multiplier?

For consumed goods: equal marginal returns

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The subjective marginal exchange rate equals the market exchange rate.
This is an interior condition, not a universal rule for every optimum.

For a good at zero: an inequality

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Why would a rule that sets every marginal return equal miss some solutions?

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Quasiconcavity alone does not make every stationary point optimal. → Sufficiency proof → Counterexample

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A global saddle gives a feasible optimum and complementary slackness. Concavity links KKT to this global maximisation. → Details

Checkpoint

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What established existence? What could establish uniqueness?
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When is a KKT solution sufficient for global optimality?
What can go wrong if we check only the interior first-order equations?

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Now an interior method is justified; it was not assumed at the start.

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The log representation is concave on the positive domain.

The bundle does not change, but the numerical utility and multiplier generally do.

Solve the interior conditions

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If $\alpha = 1/2$, $p = (2, 1)$ and $m = 12$, demand is $(3, 6)$.

Which of these predictions is a modelling restriction rather than an accounting identity?

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Keep $\alpha = 1/2$ and $m = 12$. Raise p_1 from 2 to 3, holding $p_2 = 1$.

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Does an unchanged quantity of good 2 mean that its utility contribution is unchanged?

Example 2: perfect substitutes

$$u(x) = ax_1 + bx_2, \qquad a, b > 0.$$

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If $a/p_1 > b/p_2$, buy $(m/p_1, 0)$. If the inequality reverses, buy $(0, m/p_2)$.
If the ratios are equal, every point on the budget line is optimal.

Verify a corner with the full KKT system

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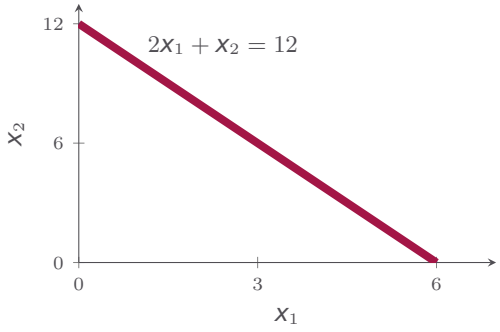
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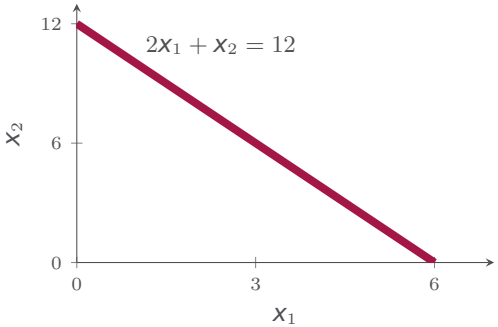
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Check: $2 - 3\lambda + \mu_1 = 0$, $1 - \lambda + \mu_2 = 0$.
The zero quantity is optimal precisely because its marginal return is too low.

Equal returns: a whole set of demand



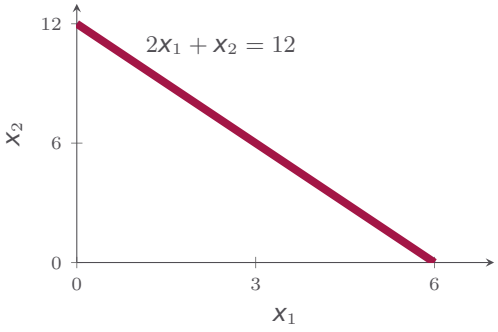
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Every budget-line bundle gives utility 12.
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At $p = (2, 1)$ and $m = 12$, the optimum is $(4, 4)$.
The optimum is a kink: the smooth stationarity formula does not apply there.

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If $m > 4$: $a^* > 0$, $\lambda = 1$, and extra wealth goes to a.
If $0 < m < 4$: $a^* = 0$, $\lambda = 2/\sqrt{m} > 1$, $\mu_a = \lambda - 1 > 0$.

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5. **Interpret:** quantities, expenditure shares and reactions to parameters.

From a classroom solution to a structural model

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Fitting one choice does not establish the functional form. Use more variation and test the restrictions.

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NES

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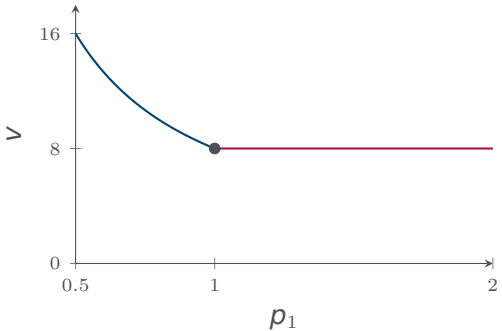
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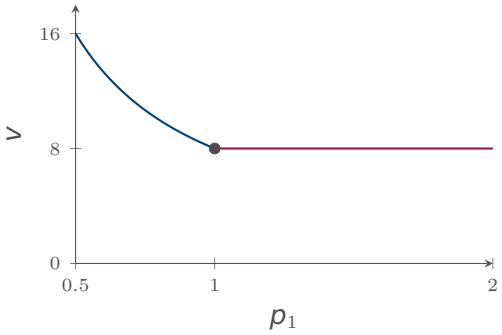
$\rightarrow$ More than two goods

A continuous value need not be differentiable



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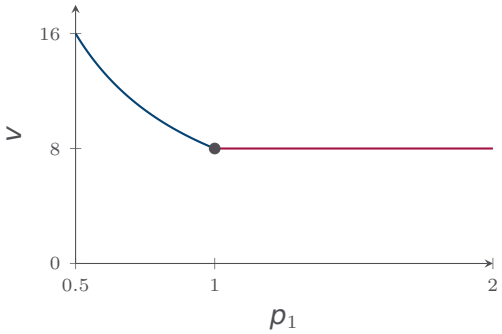
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At $p_1 = 1$, the consumer is indifferent along the budget line. The value has a kink.

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We need appropriate regularity and a differentiable value; a kink is a warning.

Differentiate the value, not the choice

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The wealth effect is positive when $\lambda^* > 0$; a higher price hurts in proportion to the quantity purchased.

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A binding budget does not by itself imply $v_m > 0$. → A counterexample

Roy’s identity: recover demand from value

Under the maintained consumer assumptions, suppose v is differentiable at (p, m) and $v_m(p, m) > 0$.

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Under these conditions the optimal quantity vector is unique. → Derivation

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Likewise $x_2 = (1-\alpha)m/p_2$. The demand agrees with the direct solution.

Demand survives a change of utility scale

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A strictly increasing transformation can have a zero derivative at a point; then the differentiable ratio may be unavailable.

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What justifies treating a KKT candidate as globally optimal?
What two differentiability/positivity conditions are needed to use Roy's formula?

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Homework 1 is issued after lecture block 3, according to the updated course schedule.

Next block: two ways to ask the consumer question

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Expenditure minimisation: given $p, \bar{u}$, how little spending attains the target?
One fixes resources; the other fixes a level of the same preferences.

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e is the expenditure function; H is the Hicksian demand correspondence. Assume the target is attainable. A singleton is denoted $h(p, \bar{u})$.

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A continuous cost attains a minimum there; excluded bundles are more expensive. → Details

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Convex preferences give a convex minimiser set; strict convexity, with the maintained conditions, gives uniqueness.

Hicksian demand: basic properties

Price homogeneity: $H(tp, \bar{u}) = H(p, \bar{u})$ for $t > 0$.

No excess utility: if $\bar{u} \geq u(0)$, every minimiser attains exactly $\bar{u}$.

Convex preferences give a convex minimiser set; strict convexity, with the maintained conditions, gives uniqueness.

No excess utility uses continuity and positive prices, not LNS.

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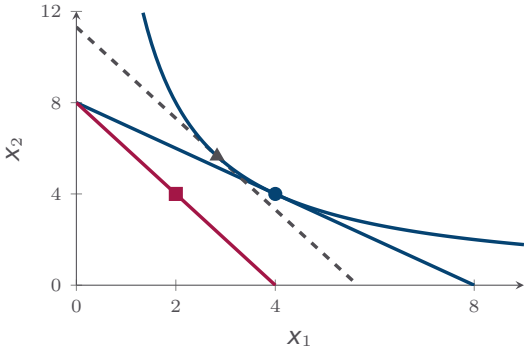
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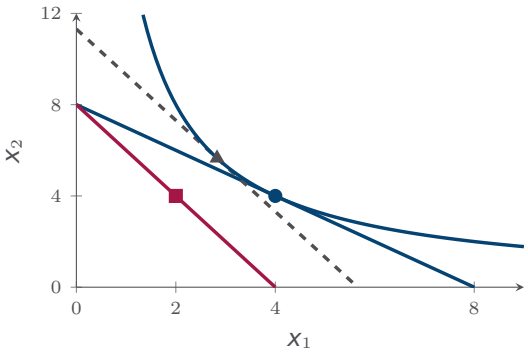
It is the infimum of linear cost functions. → Proofs

Fixed wealth versus fixed utility



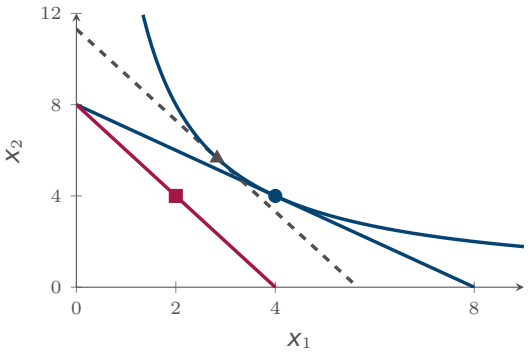
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Raise p_1 to 2: Marshallian demand at the same wealth is $(2, 4)$.
To keep utility 4, spend $8\sqrt{2}$ and buy $(2\sqrt{2}, 4\sqrt{2})$.

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These connect the two problems; they are not an assumption that the consumer literally solves both.

A monetary compensation question

Let $u^0 = v(p^0, m)$. How much extra wealth restores this utility after a price change to p^1?

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A money-metric change is useful, but summing it across people still involves welfare judgements.

Proof: LNS and budget exhaustion

Suppose an optimum x^* has slack $\delta = m - p \cdot x^* > 0$.

→ Back to budget exhaustion

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If $\|y - x^*\| < \delta / \|p\|$, then

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LNS supplies a strictly better such y in the consumption domain, contradicting optimality.

→ Back to budget exhaustion

Proof: uniqueness from strict convexity

Suppose distinct $x, y \in D(p, m)$. Their midpoint z is feasible because the budget is convex.

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This contradicts x being optimal. There is at most one optimum; existence gives a unique one.

→ Back to uniqueness

Proof: indirect utility is quasiconvex

Write $(p^t, m^t) = t(p, m) + (1 - t)(q, n)$ with $0 < t < 1$.

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Thus every bundle affordable at the mixture is affordable at at least one endpoint.
Its utility cannot exceed $\max\{v(p, m), v(q, n)\}$. Maximise over the mixed budget.

→ Back to the property

Why the constraint qualification holds here

Active nonnegativity constraints have gradients e_i for the coordinates at zero.

→ Back to necessity and sufficiency

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The active gradients are therefore linearly independent. This includes corner optima.

→ Back to necessity and sufficiency

Proof: why concavity makes KKT sufficient

For feasible y, concavity and differentiability imply

$$u(y) - u(x^*) \leq \nabla u(x^*) \cdot (y - x^*).$$

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Thus $u(y) \leq u(x^*)$ for every feasible y.

→ Back

Proof: a global saddle is an optimum

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→ Back to saddle points

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For every feasible y,

$$u(y) \leq \mathcal{L}(y, \lambda^*, \mu^*) \leq \mathcal{L}(x^*, \lambda^*, \mu^*) = u(x^*).$$

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Thus x^* is optimal.

→ Back to saddle points

A KKT point that is not optimal

Let $u(x) = (x-1)^3$, $p = 1$, $m = 2$, $x \geq 0$.

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The function is strictly increasing and quasiconcave. That alone does not make stationarity sufficient.

→ Back

A binding budget with zero marginal value

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Roy's ratio becomes $0/0$ at that point, although demand $x = m/p$ is perfectly well-defined.

→ Back to the multiplier

Cobb–Douglas with any positive exponent sum

Let $u(x) = \prod_{i=1}^{L} x_i^{a_i}$ with $a_i > 0$ and $A = \sum_i a_i$.

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No restriction $A \leq 1$ is needed for this preference solution.

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If $v_m > 0$, every optimiser satisfies the same Roy formula, so demand is unique.

→ Back

Proof: expenditure minimisation has a solution

Choose an attainable target and a feasible $\widehat{x}$. Let

$$S = \{x \geq 0 : u(x) \geq \bar{u}\} \cap \{x : p \cdot x \leq p \cdot \widehat{x}\}.$$

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S is nonempty and compact when u is continuous and $p \gg 0$.
Cost is continuous, so it has a minimum on S. Every omitted feasible bundle costs more than $\widehat{x}$, so this is a global minimum.

→ Back to the preview

Proof: homogeneity and price concavity of expenditure

NES

Scaling prices multiplies every feasible cost by the same positive number:

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$$e(p^t, \bar{u}) = tp \cdot x^t + (1-t)q \cdot x^t \geq te(p, \bar{u}) + (1-t)e(q, \bar{u}).$$

→ Back to expenditure properties